CU12 Protocol and Introduction

KERONG Industry (HK) Co., Ltd.

Electronic Research and Development Department

Friday, July 22, 2022

CONTENTS

Documentation Version 3
Overview 4
1.Introduction to CU12 Communication Protocol 5
1.1 Hardware Protocol 5
1.1.1 RS485 Protocol 5
1.2 Software Protocol 5
1.2.1 Communication Protocol Packet from Software to CU12 5
1.3 CU12 Protocol Commands Sheet(from software to CU12) 6
1.4 CU12 Protocol Response Sheet 6
1.5 Description of Commands 7
1.5.1 Get Status 7
1.5.2 Unlock 7
1.5.3 Query and Set Unlocking Time 8
1.5.4 Query and Set Baud Rate 9
1.5.5 Query and Set the Delayed Unlocking Time 10
1.5.6 Query and Set Waiting Time of Push Door Unlocking 11
1.5.7 Initialization 12
1.5.8 Query the Version Information 13
2.Introduction to CU12 Hardware 14
2.1 Parameters 14
2.2 Description of Hardware 15
2.2.1 Connecting Port Description 16
2.2.2 Address Description 16
2.2.3 Button Description 16
2.2.4 LED Description 16
2.2.5 RS485 Connecting Description 16
3.CU12 Connection 17
3.1 RS485 Connection 17
3.2 TCP/IP Connection 20
4.Installation Instruction 31

Documentation Version

Date	Version	Author	Revised content
Wednesday, 06 July, 2022	V1.0	Wang Yonglin	Create the documentation
Friday, 22 July, 2022	V1.1	Chen Shiwei	Add Hardware Description, PCB board diagram

Overview

This instruction introduces the CU12 Board communication protocol in detail.

Section 1: CU12 communication protocol instructions;

Section 2: CU12 hardware description.

Section 3: CU12 connection

1.Introduction to CU12 Communication Protocol

1.1 Hardware Protocol

1.1.1 RS485 Protocol

Items	Description
Baud rate	19200 (Factory default, can be modified)
Data bites	8 Bytes
Check bites	None
Stop bites	1 Byte

1.2 Software Protocol

1.2.1 Communication Protocol Packet from Software to CU12

The maximum length of a protocol packet is 48 bytes

The minimum length of a protocol packet is 8 bytes

Protocol package format: Big-endian

General format is as follows:

NO.	Items	Description
1	STX(1byte)	Start code(fixed value): 0x02
2	ADDR(1byte)	ADDR of CU12 hardware: 0x00~0x10 Eg.: ADDR=0x00, means to control CU12 which ADDR is 0x00 Notice: ADDR=0x10, means send commands to all slaves of the bus
3	LOCKNUM(1byte)	Lock number of CU12: 0x00~0x0C. Eg.: LOCKNUM=0x00, means unlock No.0x01 lock Notice: LOCKNUM=0x0C, means unlock all 12pcs locks of CU12
4	CMD(1byte)	Command, refers to CU12 protocol commands sheet
5	ASK(1byte)	Response value, refers to the CU12 protocol response sheet
6	DATALEN(1byte)	Length of data DATA
7	ETX(1byte)	End code(fixed value): 0x03
8	SUM(1byte)	The low byte of the whole command packet's checksum. Eg: SUM=STX+ADDR+LOCKNUM+CMD+ASK+DATALEN+ETX+DATA=0x125D , so SUM=0x5D
... 8+DATELEN	DATA (DATALEN bytes)	Data, when DATALEN is zero, there is no this data. Different command data formats are different. Refer to the detailed instructions.

1.3 CU12 Protocol Commands Sheet(from software to CU12)

NO.	Command	Code	Description
Communication command (Software <——> CU12)			
1	Get status	0x80	Get the status of the lock
2	Unlock	0x81	Unlock command
3	Query and set the unlocking time	0x82	Query and set the unlocking time
4	Query and set baud rate	0x83	Query and set baud rate of CU12
5	Query and set the delayed time of unlocking	0x84	Query and set CU12 delayed time of unlocking
6	Query and set waiting time of push door unlocking	0x85	Query and set CU12 waiting time of push door unlocking
7	Initialization	0x8E	Initialize CU12
8	Querying version information	0x8F	Query the CU12 version information
Note			

1.4 CU12 Protocol Response Sheet

ASK value	Description
When data is returned, the meaning of response value	
0x10	Correct operation
0x11	Failed Operation
0x12	Wait timeout
0x13	Unknown command
0x14	Data verification failure
When a command is actively sent, the meaning of this value	
0x00	Default value for ASK when sending a command

1.5 Description of Commands

Detailed description of commands for customers' understanding.

1.5.1 Get Status

(1) Protocol

Command: 0x80

(2) Command function

Get the CU12's lock hook status

(3) Format of data Data

u8 HookStateDatas[2];

(4) Examples

a) Get status of specific CU12

(the following is to get status of CU12 which ADDR is 0x00):

Command (software——>CU12) : 02 00 00 80 00 00 03 85

Response (CU12—>software) : 02 00 00 80 10 02 03 99 02 00

b) Get status of all slaves of the bus

(the following is to get status of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus):

Command (software——>CU12) : 02 10 00 80 00 00 03 95

Response (CU12——>software) : 02 00 00 80 10 02 03 99 02 00

02 01 00 80 10 02 03 9C 04 00

(5) Explanation of examples

Example a: to get hook status of CU12 which ADDR is 0x00, 02 00 means response data of hook status. From the data, the lock No. 0x02 of CU12 which ADDR is 0x00 is locked, and the other 11 locks are unlocked.

Example b: to get the hook status of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus, 02 00 means response hook status data of CU12 which ADDR is 0x00 , 04 00 means response data of hook status of CU12 which ADDR is 0x01. From the data we can know, the lock No.0x02 of CU12 which ADDR is 0x00 is locked, and the other 11 locks are not locked. The lock No.0x03 of CU12 which ADDR is 0x01 is locked, and the others are unlocked.

1.5.2 Unlock

(1) Protocol

Command: 0x81

(2) Command Function

Unlock the specific lock of CU12 according to the specific command;

(3) Format of data DATA:

None

(4) Examples

a) Unlock the specific lock on specific CU12

(the following is to unlock the lock No.3 on CU12 which ADDR is 0x00) :

Command (software——>CU12) : 02 00 02 81 00 00 03 88

Response (CU12——>software) : 02 00 02 81 10 00 03 98

b) Unlock all 12 locks on specific CU12

(the following is to unlock all locks on CU12 which ADDR is 0x00) :

Command (software——>CU12) : 02 00 0C 81 00 00 03 92

Response (CU12——>software) : 02 00 0C 81 10 00 03 A2

c) Unlock all CU12 locks on the RS485 bus

(the following is to unlock all 12 locks of the all CU12 on the RS485 bus)

Command (software——>CU48) : 02 10 0C 81 00 00 03 A2

Response (CU48——>software) : 02 00 0C 81 10 00 03 A2

02 01 0C 81 10 00 03 A3

(4) Explanation of examples

Example a: ADDR=0x00 and LOCKNUM=0x02 indicate that unlock the third lock on CU12 No. 0x00

Example b: ADDR=0x00, LOCKNUM=0x0C indicate that unlock all 12 locks on CU12 NO.0x00

Example c: ADDR=0x10, LOCKNUM=0x0C indicate that unlock all slaves of the bus.

1.5.3 Query and Set Unlocking Time

(1) Protocol

Command: 0x82

(2) Command Function

Query and set the lock unlocking time of CU12. The value of lock unlocking time ranges from 0 to 6000 (decimal), indicating 0 ms-60000ms. The default value is 55 (in decimal notation), which indicates 550ms.

(3) Format of data DATA:

u16 OpenLockTime;

(4) Examples:

a) Query the unlocking time of specific CU12

(the following is to query the unlocking time of CU12 which ADDR is 0x00)

Command (software——>CU12) : 02 00 00 82 00 00 03 87

Response (CU12——>software) : 02 00 00 82 10 02 03 D0 00 37

b) Query the lock opening time of all CU12 on the RS485 bus

(the following is to query the lock opening time of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus) :

Command (software——>CU12) : 02 10 00 82 00 00 03 97

Response (CU12——>software) : 02 00 00 82 10 02 03 D0 00 37

02 01 00 82 10 02 03 D1 00 37

c) Set the unlocking time of specific CU12

(the following is to set the unlocking time of CU12 which ADDR is 0x00) :

Command (software——>CU12) :02 00 00 82 00 02 03 ED 00 64

Response (CU12——>software) :02 00 00 82 10 00 03 97

d) Set the unlocking time of all CU12 on the RS485 bus

(the following is to set the unlocking time of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus) :

Command (software——>CU12) :02 10 00 82 00 02 03 FD 00 64

Response (CU12——>software) : 02 00 00 82 10 00 03 97

02 01 00 82 10 00 03 98

(5) Explanation of Examples

Example a: to query the unlocking time of CU12 which ADDR is 0x00, where 00 37 indicates the unlocking time data. According to the data, the unlocking time of CU12 which ADDR is 0x00 =0x0037x10ms=550ms;

Example b: to query the unlocking time of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus, where 00 37 indicates that the unlocking time data of CU12 which ADDR is 0x00 and 0x01 are obtained. According to the data we can know, the unlocking time of CU12 which ADDR is 0x00 and 0x01=0x0037x10ms=550ms;

Example c: to set the unlocking time of CU12 which ADDR is 0x00, where 00 64 indicates the unlocking time data to be set. According to the data: the unlocking time of CU12 which ADDR is 0x00 =0x0064x10ms=1000ms;

Example d: to set the unlocking time of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus, where 00 64 indicates setting the unlocking time data of CU12 which ADDR is 0x00 and 0x01. According to the data we can know, unlocking time of CU12 Which ADDR is 0x00 and 0x01 =0x0064x10ms=1000ms;

Notice: The default unlocking time after initialization is 0x0037x10ms=550ms. After sending the command to set the unlocking time, CU12 will enter the data saving task.

1.5.4 Query and Set Baud Rate

(1) Protocol

Command: 0x83

(2) Command Function

Query and set the baud rate of CU12, the value range of the baud rate parameters is 0-3 (decimal), where 0 = 9600, 1 = 19200 (factory default value) , 2 = 57600, 3 = 115200;

(3) Format of data DATA:

u8 BaudRateFlag;

(4) Examples

a) Query baud rate of specific CU12

(the following is to query baud rate of CU12 which ADDR is 0x00):

Command (software——>CU12) : 02 00 00 83 00 00 03 88

Response (CU12——>software) : 02 00 00 83 10 01 03 9A 01

b) Query the baud rate of all CU12 on the RS485 bus

(the following is to query the baud rate of CU12 which ADDR is 0x00 and address 0x01 on the RS485 bus) :

Command (software——>CU12) : 02 10 00 83 00 00 03 98

Response (CU12——>software) : 02 00 00 83 10 01 03 9A 01

02 01 00 83 10 01 03 9B 01

c) Set the baud rate for specific CU12

(the following is to set the baud rate for CU12 which ADDR is 0x00) :

Command (software——>CU12) : 02 00 00 83 00 01 03 89 00

Response (CU12——>software) : 02 00 00 83 10 00 03 98

d) Set the baud rate of all CU12 on the RS485 bus

(the following is to set the baud rate of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus) :

Command (software——>CU12) : 02 10 00 83 00 01 03 99 00

Response (CU12——>software) : 02 00 00 83 10 00 03 98

02 01 00 83 10 00 03 99

(5) Explanation of Examples

Example a: to query the baud rate of CU12 which ADDR is 0x00, and 01 indicates the baud rate. According to the data, the baud rate of CU12 which ADDR is 0x00=19200;

Example b: to query the baud rate of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus, 01 indicates that the baud rate of CU12 which ADDR is 0x00 and 0x01 is obtained. According to the data, the baud rate of CU12 which ADDR is 0x00 and 0x01=19200.

Example c: to set the baud rate of CU12 which ADDR is 0x00, where 00 indicates the baud rate to be set. According to the data, the baud rate of CU12 which ADDR is 0x00 =9600;

Example d: to set the baud rate of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus. 00 indicates setting the baud rate of CU12 which ADDR is 0x00 and 0x01. According to the data, the baud rate of CU12 which ADDR is 0x00 and 0x01=9600;

Notice: After initialization, the default baud rate is 19200. After sending the command to set the baud rate, CU12 will enter the data saving task.

1.5.5 Query and Set the Delayed Unlocking Time

(1) Protocol

Command: 0x84

(2) Command Function

Query and set the CU12 delayed unlocking time. The value ranges from 0 to 200 (in decimal notation), the unit of time is second. The default value is 0, indicating that the CU12 is unlocked without delay.

(3) Format of data DATA:

u8 DelayOpenLockTIme;

(4) Examples:

a) Query the delayed unlocking time of specific CU12

(the following is to query the delayed unlocking time of CU12 which ADDR is 0x00) :

Command(software——>CU12) : 02 00 00 84 00 00 03 89

Response(CU12——>software) : 02 00 00 84 10 01 03 9A 00

b) Query the delayed unlocking time of all CU12 on the RS485 bus:

(the following is to query the delayed unlocking time of CU12 which ADDR is 0x00 and 0x01)

Command (software——>CU12) : 02 10 00 84 00 00 03 99

Response (CU12——>software) : 02 00 00 84 10 01 03 9A 00

02 01 00 84 10 01 03 9B 00

c) Set the delayed unlocking time of specific CU12

(the following is to set the delayed unlocking time of CU12 which ADDR is 0x00) :

Command (software——>CU12) : 02 00 00 84 00 01 03 8A 00

Response (CU12——>software) : 02 00 00 84 10 00 03 99

d) Set the delayed unlocking time of all CU12 on the RS485 bus

(the following is to set the delayed unlocking time of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus) :

Command (software——>CU12) : 02 10 00 84 00 01 03 9A 00

Response (CU12——>software) : 02 00 00 84 10 00 03 99

02 01 00 84 10 00 03 9A

(5) Explanation of examples

Example a: to the query the delayed unlocking time of CU12 which ADDR is 0x00, where 00 indicates the delayed unlocking time. According to the data we can know, the delayed unlocking time of CU12 which ADDR is 0x00 is 0, indicating that the lock is directly unlocked without delay;

Example b: to query the delayed unlocking time of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus, where 00 means the delayed unlocking time of CU12 which ADDR is 0x00 and 0x01 is obtained. According to the data we can know, the delayed unlocking time of CU12 which ADDR is 0x00 and 0x01 is 0, indicating that the CU12 is unlocked without delay;

Example c: to set delayed unlocking time of CU12 which ADDR is 0x00, where 00 indicates the delayed unlocking time to be set. According to the data we can know, the delayed unlocking time of CU12 which ADDR is is 0x00 is 0, indicating the direct unlocking without delay;

Example d: to set the delayed unlocking time of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus, where 00 indicates setting the delayed unlocking time of CU12 which ADDR is 0x00 and 0x01. According to the data we can know: The delayed unlocking time of CU12 which ADDR is 0x00 and 0x01 is 0, indicating that the CU12 is unlocked without delay;

Notice: The default value of delay unlocking time after initialization is 0. In particular, after sending the command of setting delay unlocking time, CU12 will enter the data saving task.

1.5.6 Query and Set Waiting Time of Push Door Unlocking

(1) Protocol

Command: 0x85

(2) Command function

Query and set CU12 waiting time of Push door unlocking. The value ranges from 0 to 200 (decimal), the unit of time is second. The default value is 0, indicating that the door is directly unlocked without pushing door.

(3) Format of data DATA:

u8 WaitPushDoorOpenLockTIme;

(4) Examples:

a) Query the waiting time of push door unlocking on specific CU12

(the following is to query the waiting time of push door unlocking of CU12 which ADDR is 0x00) :

Command (software——>CU12) : 02 00 00 85 00 00 03 8A

Response (CU12——>Software) : 02 00 00 85 10 01 03 9B 00

b) Query the waiting time of push door unlocking of all CU12 on the RS485 bus

(the following is to query waiting time of push door unlocking of CU12 which ADDR is 0x00 and 0x01) :

Command (software——>CU12) : 02 10 00 85 00 00 03 9A

Response (CU12——>Software) : 02 00 00 85 10 01 03 9B 00

02 01 00 85 10 01 03 9C 00

c) Set the waiting time of push door unlocking on specific CU12

(the following is to set the waiting time of push door unlocking on CU12 which ADDR is 0x00) :

Command (software——>CU12) : 02 00 00 85 00 01 03 8B 00

Response (CU12——>Software) : 02 00 00 85 10 00 03 9A

d) Set the waiting time of Push door unlocking of all CU12 on the RS485 bus

(the following is to set waiting time of push door unlocking of CU12 which ADDR is 0x00 and 0x01) :

Command (software——>CU12) : 02 10 00 85 00 01 03 9B 00

Response (CU12——>Software) : 02 00 00 85 10 00 03 9A

02 01 00 85 10 00 03 9B

(5) Explanation of examples

Example a: to query the waiting time of Push door unlocking of CU12 which ADDR is 0x00, where 00 indicates the waiting time of push door unlocking. According to the data: the waiting time of push door unlocking of CU12 which ADDR is 0x00 is 0, indicating that the door is directly unlocked without pushing the door.

Example b: to query the waiting time of push door unlocking of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus, where 00 indicates to obtain the waiting time of Push door unlocking of CU12 which ADDR is 0x00 and 0x01. According to the data, the waiting time of push door unlocking of CU12 which ADDR is 0x00 and 0x01 is 0, indicating that the door is directly unlocked without pushing the door;

Example c: to set the waiting time of push door unlocking of CU12 which ADDR is 0x00, where 00 indicates the waiting time of push door unlocking of CU12 which ADDR is 0x00. According to the data, the waiting time of push door unlocking of CU12 which ADDR is 0x00 is 0, indicating that the door is directly unlocked without pushing the door;

Example d: to set the waiting time of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus, where 00 indicates to set the waiting time of push door unlocking of CU12 which ADDR is 0x00 and 0x01. It can be learned from the data that: The waiting time of push door unlocking of CU12 which ADDR is 0x00 and 0x01 is 0, indicating that the lock is directly unlocked without pushing the door;

Notice: After initialization, the default waiting time for the door to be pushed and unlocked is 0. In particular, after sending the command of setting the time of waiting for the door to be pushed and unlocked, CU12 will enter the data saving task. It is recommended to wait for 500ms or more at this time, and then send the next command;

Before setting the waiting time for push door unlocking to 1-200, it is necessary to confirm that the logistics cabinet lock in use has the function of push door unlocking detection, otherwise it cannot be used normally. Once the function of waiting for door opening is enabled, the function of delayed door opening will be prohibited.

1.5.7 Initialization

(1) Protocol

Command: 0x8E

(2) Command Function

Initialize the parameters of CU12

(3) Format of data DATA:

None

(4) Examples:

a) Initialize the status of specific CU12

(the following is to initialize CU12 which ADDR is 0x00) :

Command (Software——>CU12) : 02 00 00 8E 00 00 03 93

Response (CU12——>Software) : 02 00 00 8E 10 00 03 A3

b) Initialize all CU12 on the RS485 bus

(the following is to initialize CU12 which ADDR is 0x00 and 0x01 on the RS485 bus) :

Command (Software——>CU12) : 02 10 00 8E 00 00 03 A3

Response (CU12——>Software) : 02 00 00 8E 10 00 03 A3

02 01 00 8E 10 00 03 A4

(5) Explanation of Examples

Example a: to initialize CU12 which ADDR is 0x00;

Example b: to initialize CU12 which ADDR is 0x00 and 0x01 on the RS485 bus;

Notice: After sending the initialization command, CU12 will start the data saving task; It is recommended to wait more than 500ms before sending next command.

1.5.8 Query the Version Information

(1) Protocol

Command: 0x8F

(2) Command Function

query the software and hardware versions of CU12.

(3) Format of data DATA:

```
struct
{
u8 softversion;
u8 hardversion;
}version;
```

(4) Examples:

a) Query the version information of specific CU12

(the following is to query the version information of CU12 which ADDR is 0x00) :

Command (Software——>CU12) : 02 00 00 8F 00 00 03 94

Response (CU12——>Software) : 02 00 00 8F 10 02 03 C8 10 12

b) Query the version information of all CU12 on the RS485 bus

(the following is to query the version information of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus)

Command (Software——>CU12) : 02 10 00 8F 00 00 03 A4

Response (CU12——>Software) : 02 00 00 8F 10 02 03 C8 10 12

02 01 00 8F 10 02 03 C9 10 12

(5) Explanation of Examples

Example a: to query the version information of CU12 which ADDR is 0x00, where 10 12 indicates the version information. According to the information, 10 12 is the version information of CU12 which ADDR is 0x00. And 10 12 indicates the software version is V1.0 and hardware version is V1.2;

Example b: to initialize the version information of CU12 which ADDR is 0x00 and 0x01 on the RS485 bus, where 10 12 indicates the version information. According to the data, the software version is V1.0 and hardware version is V1.2.

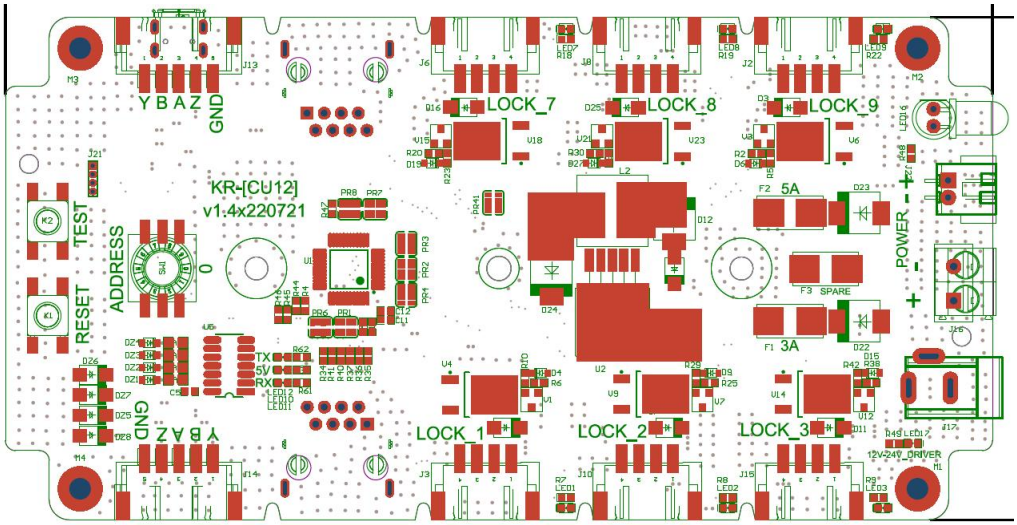
六六

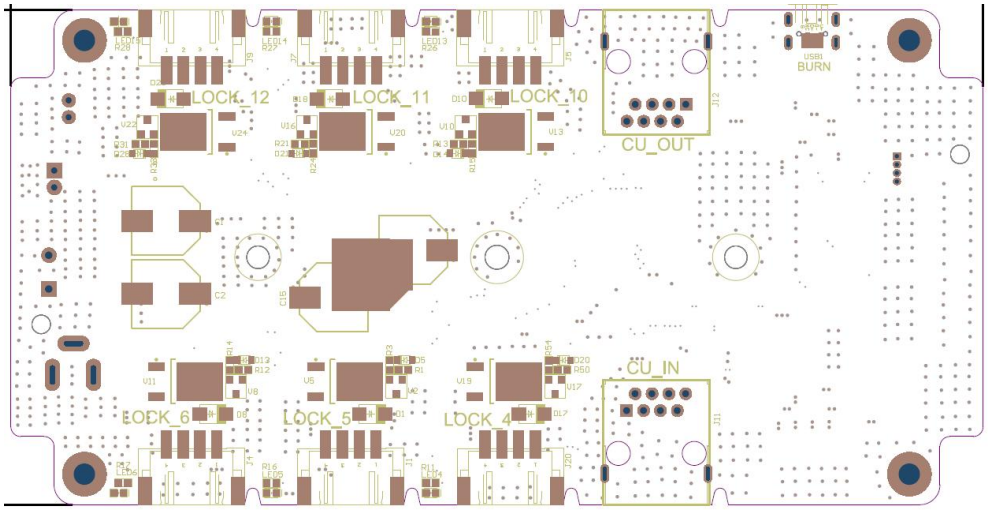
2.Introduction to CU12 Hardware

2.1 Parameters

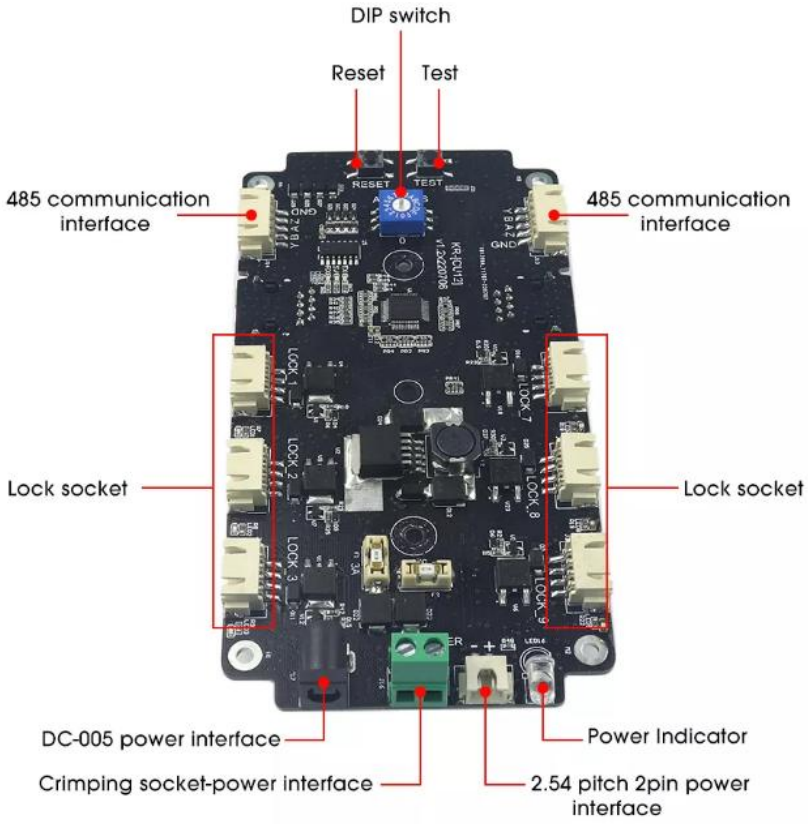
CU12 Parameters				
Working voltage	Stand-by power consumption	Drive Capability	Unlocking time	Series connection Number
DC 8~24V	20mA@12V	50W for each port	0~60000ms, the default is 500ms (time can be designed according to different locks)	Connect at most 16pcs CU12s

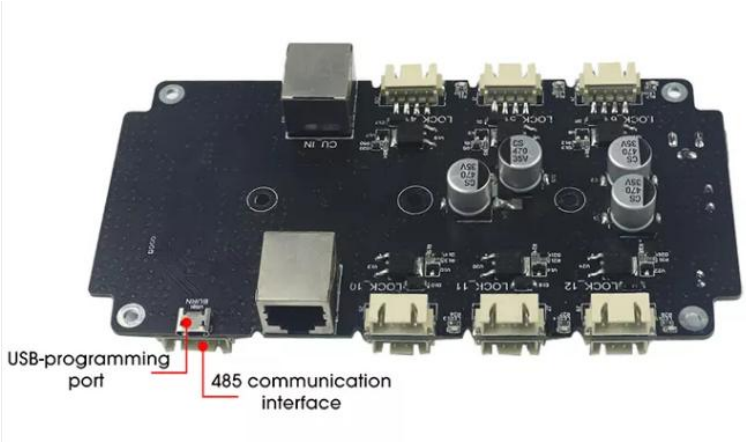
Board Diagram





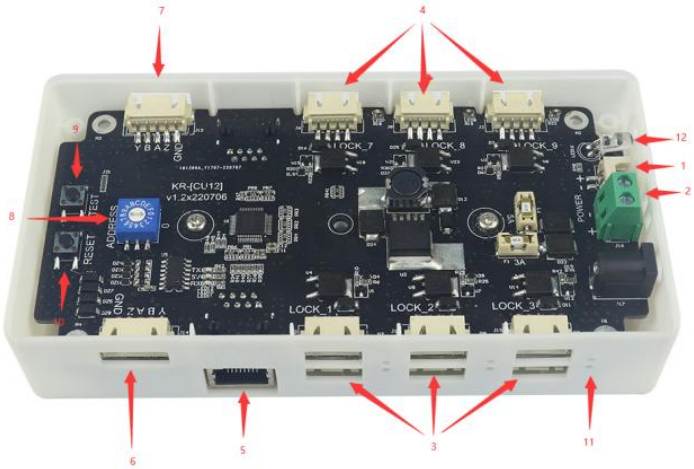
2.2 Description of Hardware





USB-Programming Port: Program upgrade or replacement can be carried out through this port;

Notice: This port cannot be used for power supply;



2.2.1 Connecting Port Description

【1】2.54 pitch 2pin power interface

【2】Crimping socket-power interface

【3】【4】 Connecting locks, supply power to locks and detect lock's condition (open/close); 4Pin connector, 2Pin for power, 2Pin for lock hook detective switch

【5】Communication for RS485, connector type is RJ45, also can for power supply, and connect another CU12.

【6】【7】Connecting port for RS485(5Pin), connector type is XH2.54.

2.2.2 Address Description

【8】Address for KR-CU12 Notice: when connect more than 1pcs CU12, their address cannot be the same

2.2.3 Button Description

Button【9】TEST, press it can operate unlocking test

Button【10】RESET

2.2.4 LED Description

LED【11】LED for hook detective. Detective switch is on, red LED off; detective switch is off, red LED on.

LED【12】To show whether power input (12V-24V), power output (5V) is normal.

2.2.5 RS485 Connecting Description

Board port ——> RS485 adaptor:

A ——> T/R+

B ——> T/R-

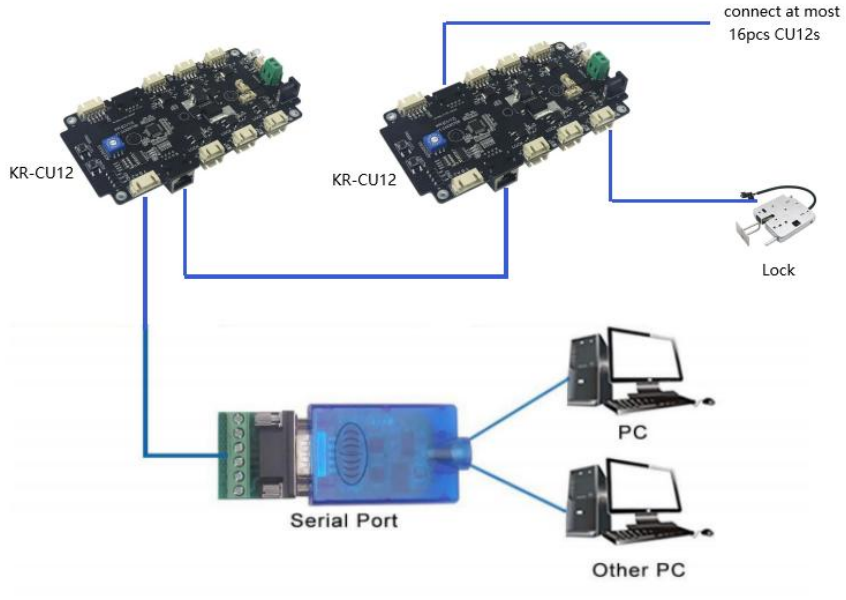
Y ——> RXD+

Z ——> RXD-

GND ——> GND

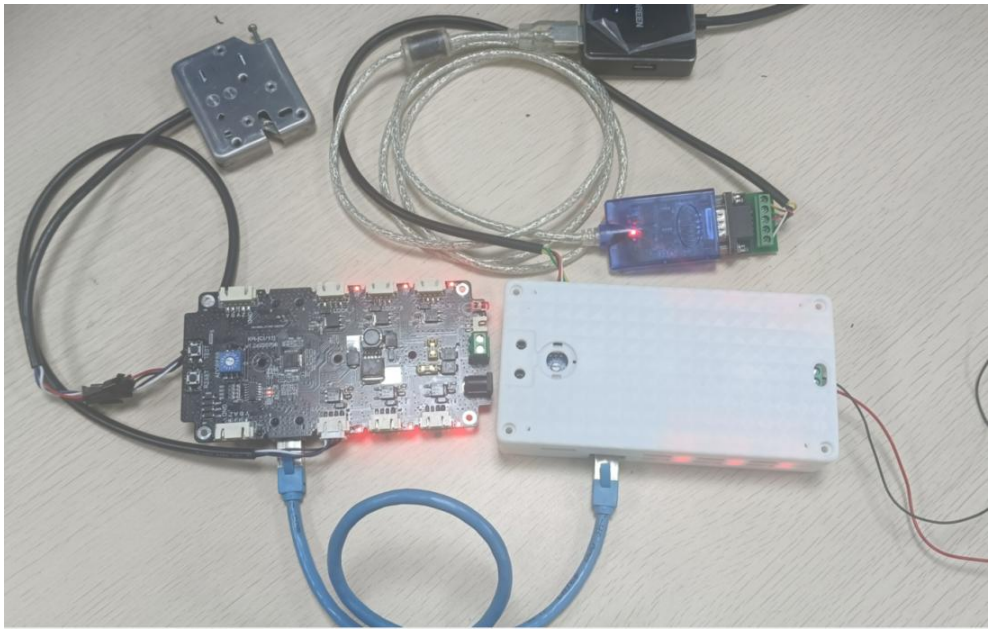
3.CU12 Connection

3.1 RS485 Connection



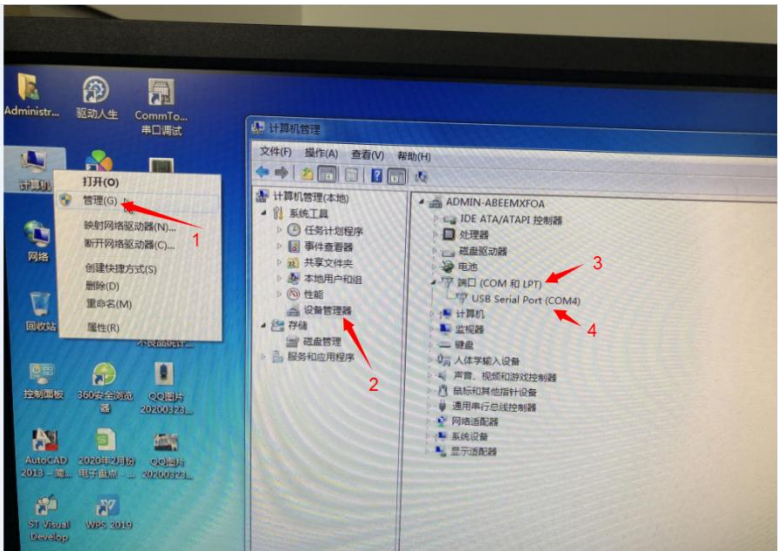
Stage 1: Connection

Correctly connect RS485, CU12 and PC. The wiring diagram is as follows:



Stage 2: Check the port number of the PC used for RS485.

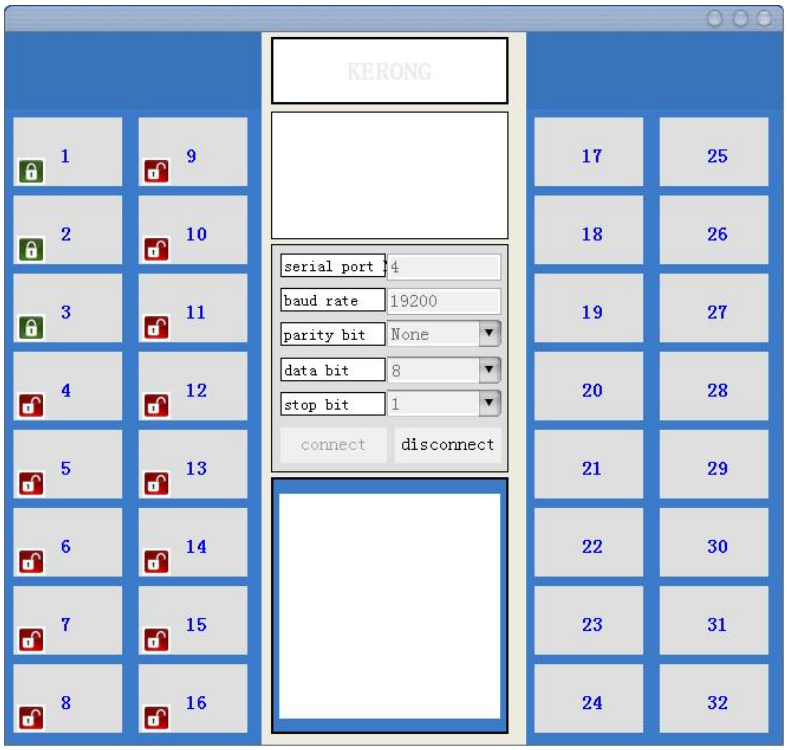
Right-click *This PC* and choose *Manage* to open the *Computer Management* interface, then select *Device Manager.* Select *Ports (COM& LPT)* in the right box to check the port number used by the RS485 port.



Stage 3 Test

(1) **Method one**

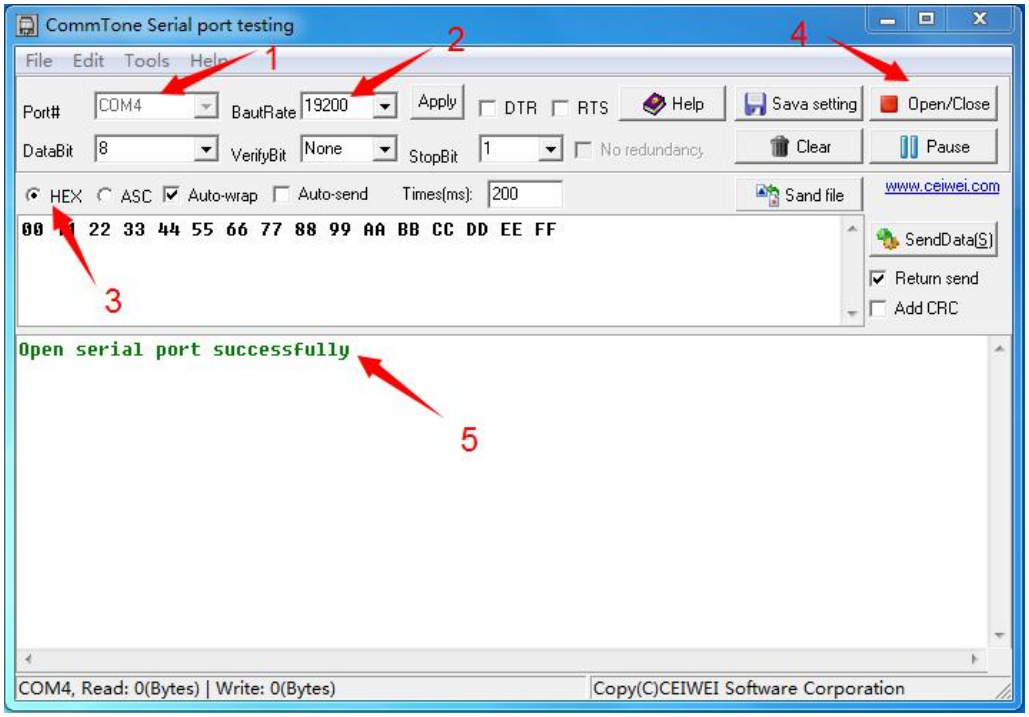
Open the RS485 test software RS485.exe， check the correct serial port(such as COM4)；baud rate: 19200；check bit: None；data bit: 8；stop bit: 1. The details are as follows:



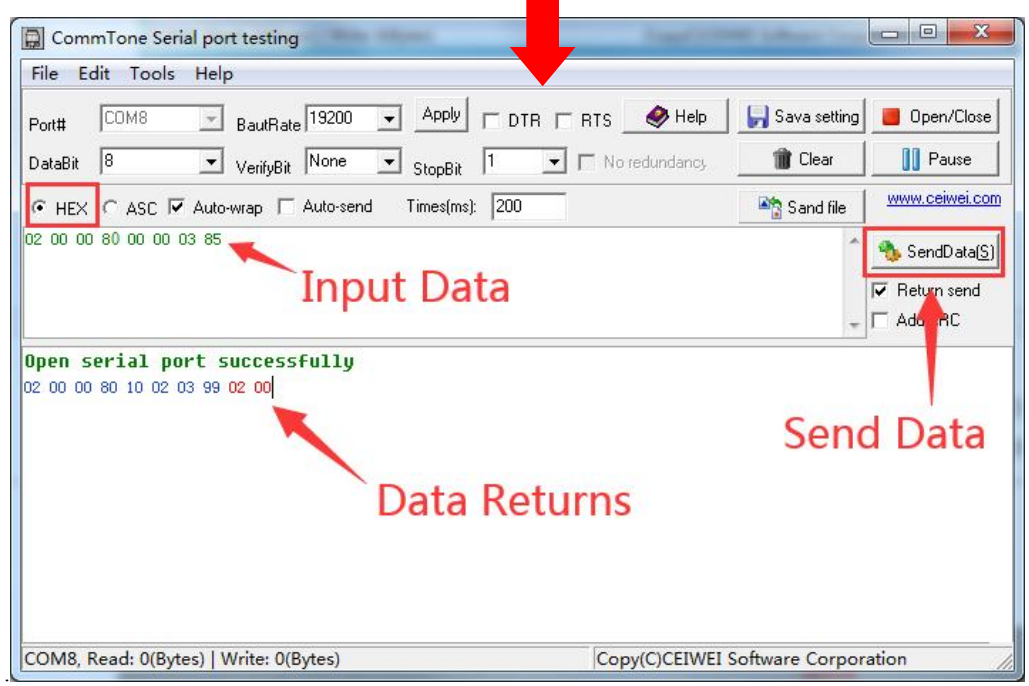
(2) Method Two:

Open the test software

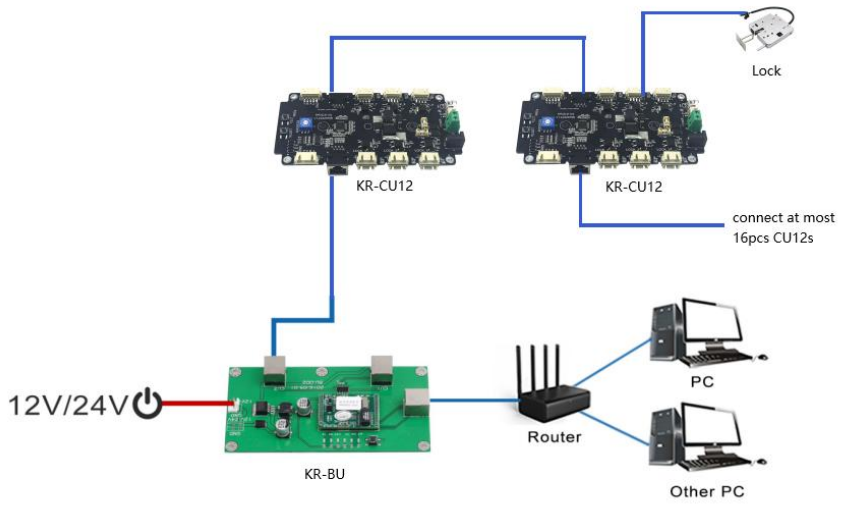
Notice: Set the baud rate to 19200 and send data in hexadecimal format (HEX), as shown in the following figure:



Use the CU12 protocol for the test. And select *HEX*



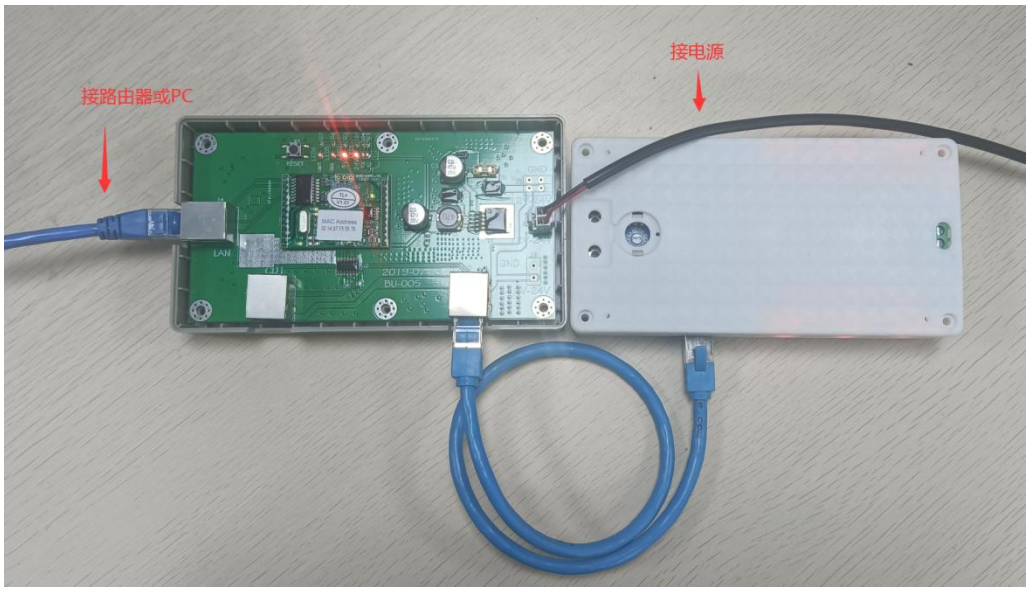
3.2 TCP/IP Connection



Stage 1 Connection

Correctly Connect BU、CU12、PC and Router.

Notice: BU should connect with router, not switcher.



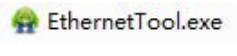
Stage 2 BU setting

Set IP address of BU.

Notice: The BU IP address cannot be the same as PC address, the last bit should be different.

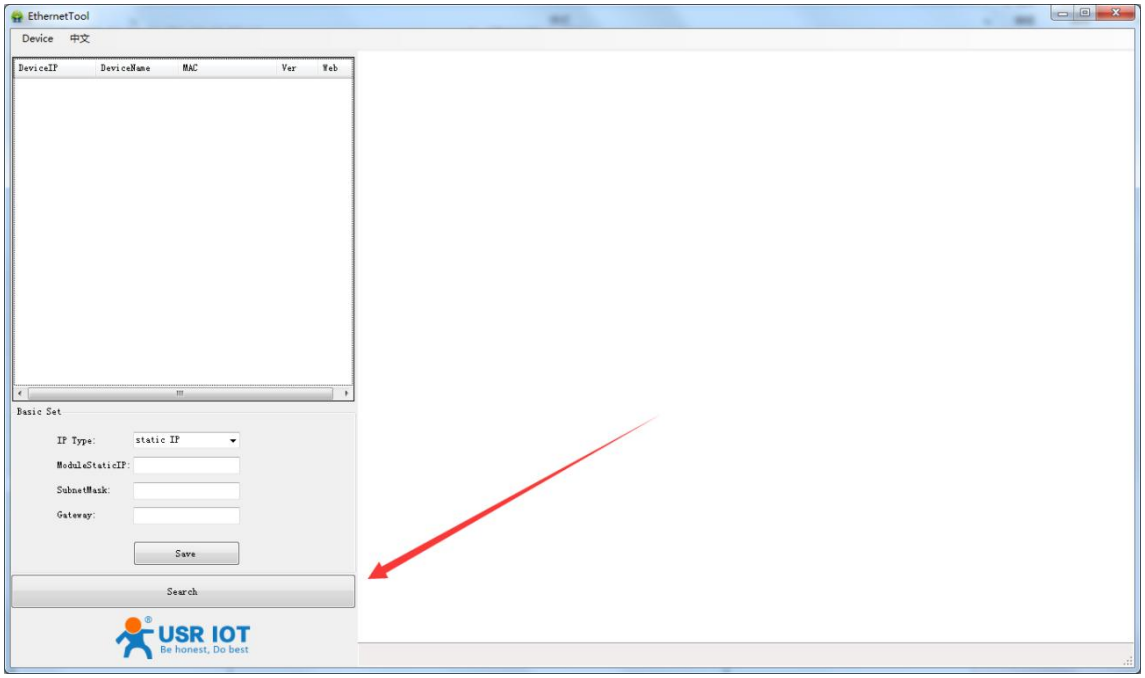
Eg.: IP address of PC is 192.168.0.100, then IP address of BU can be set as 192.168.0.158, so last bit is different.

Specific operations are as follows:

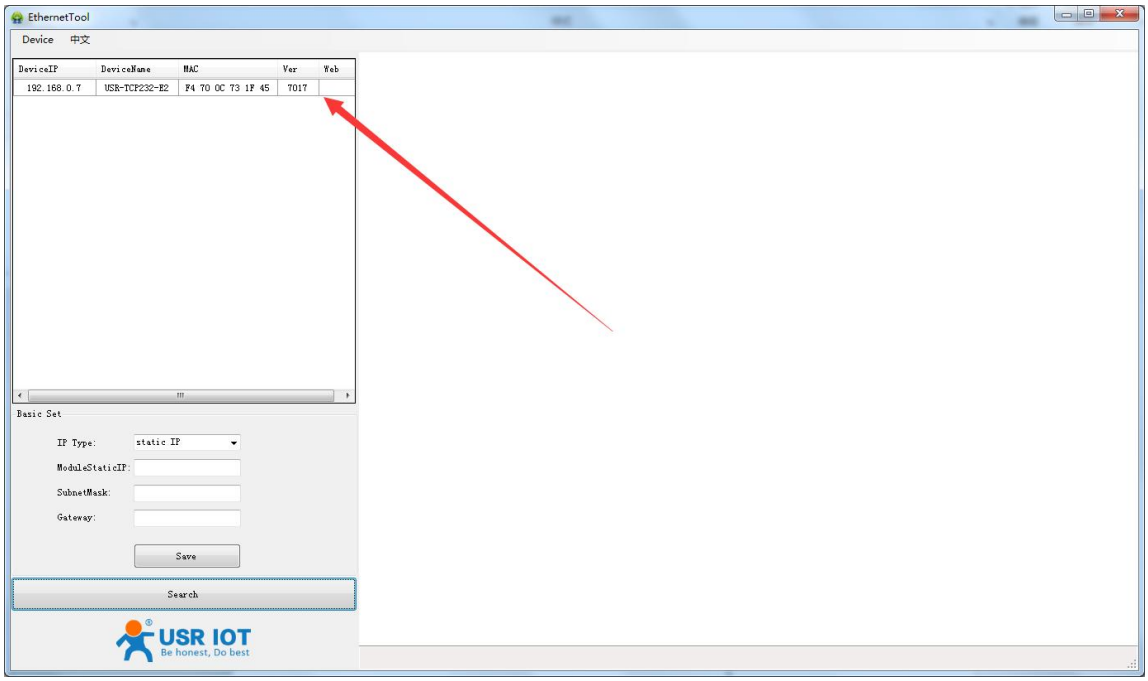
Open the BU setting software 

, and the operation is as shown below

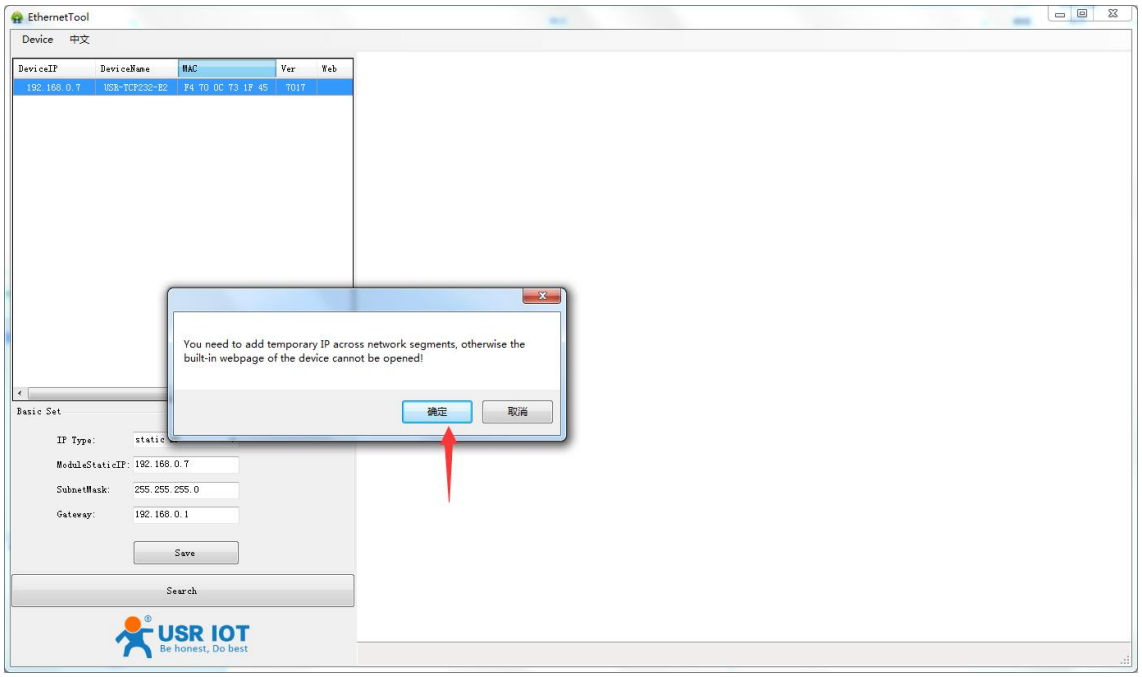
(1) Click *Search;*



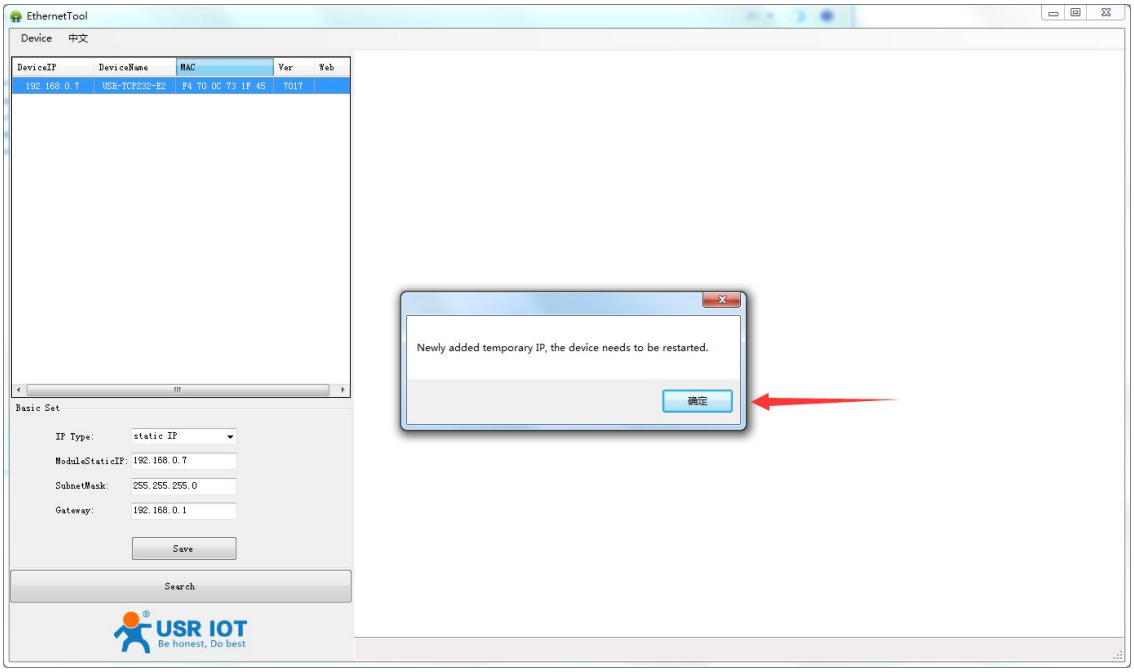
(2) Double-click the *IP address of the BU module;*



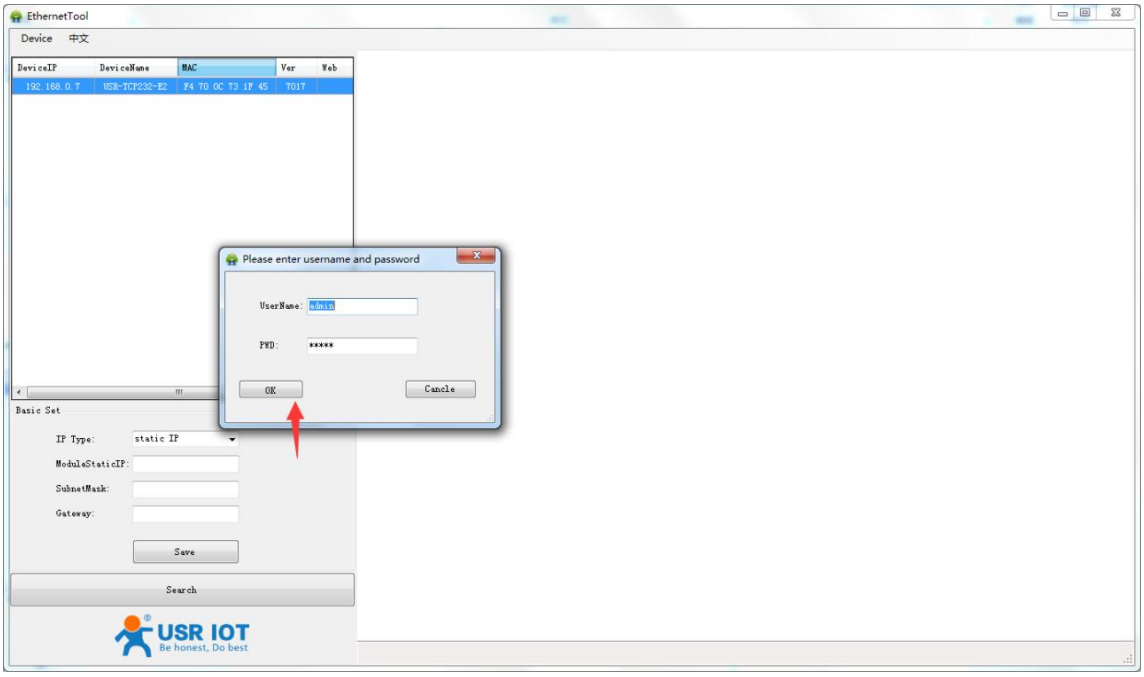
(3) Click *OK* to allow the temporary IP to be added;



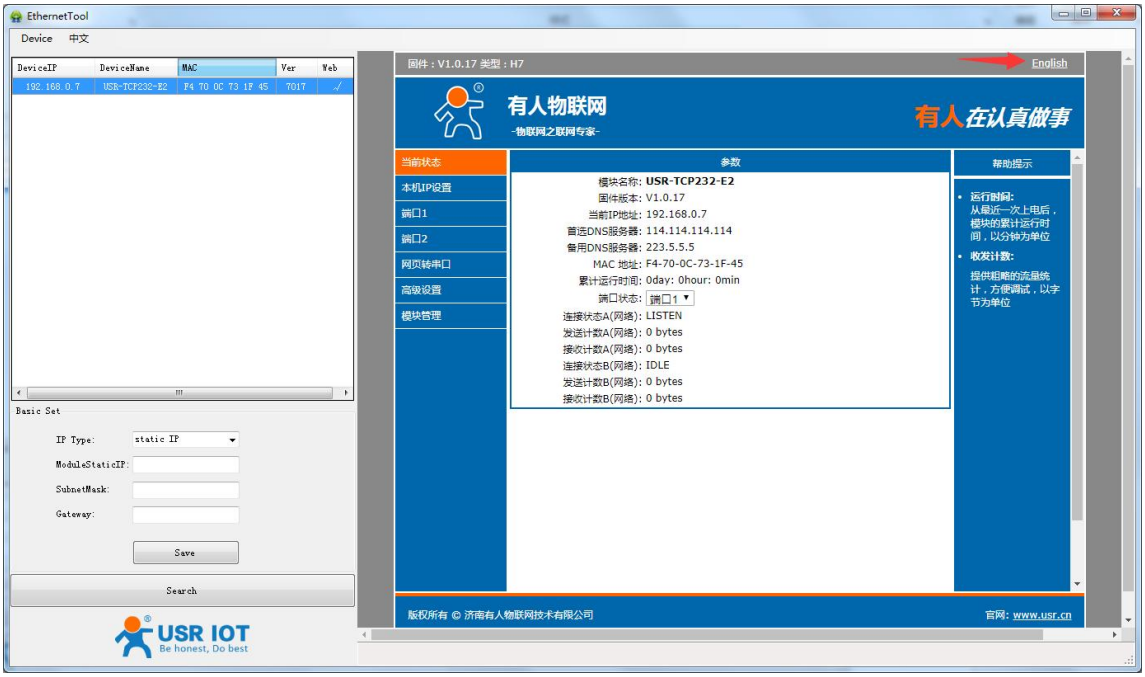
(4) Click *OK* to allow the BU to add a temporary IP address and restart;



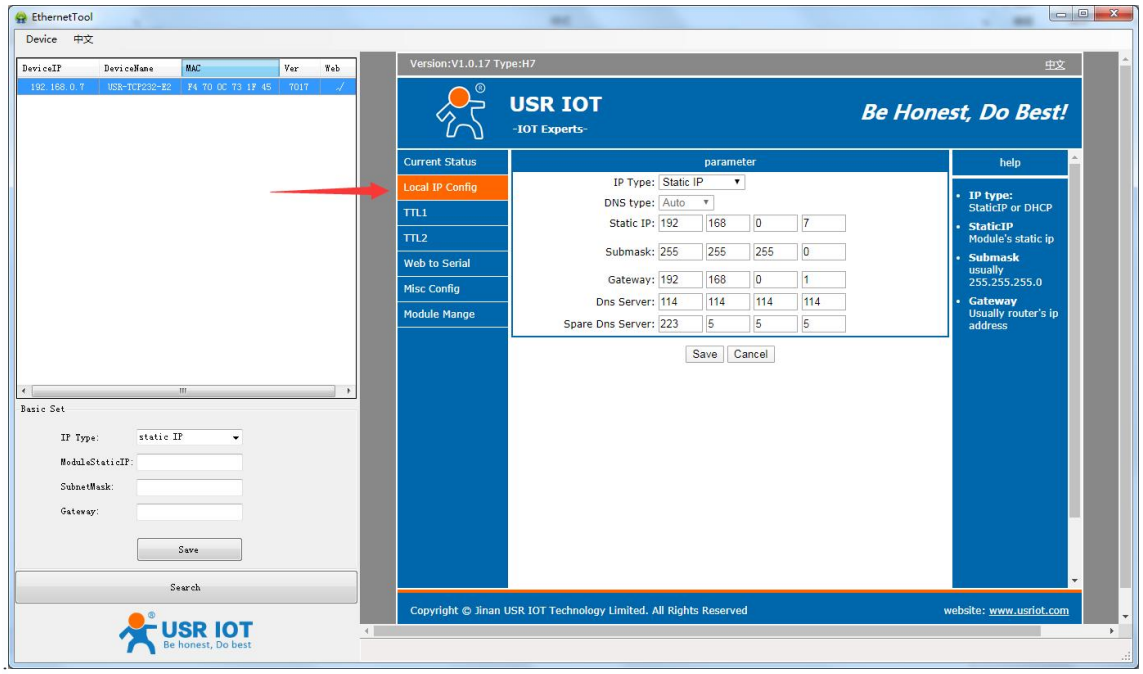
(5) Click *OK* to log in;



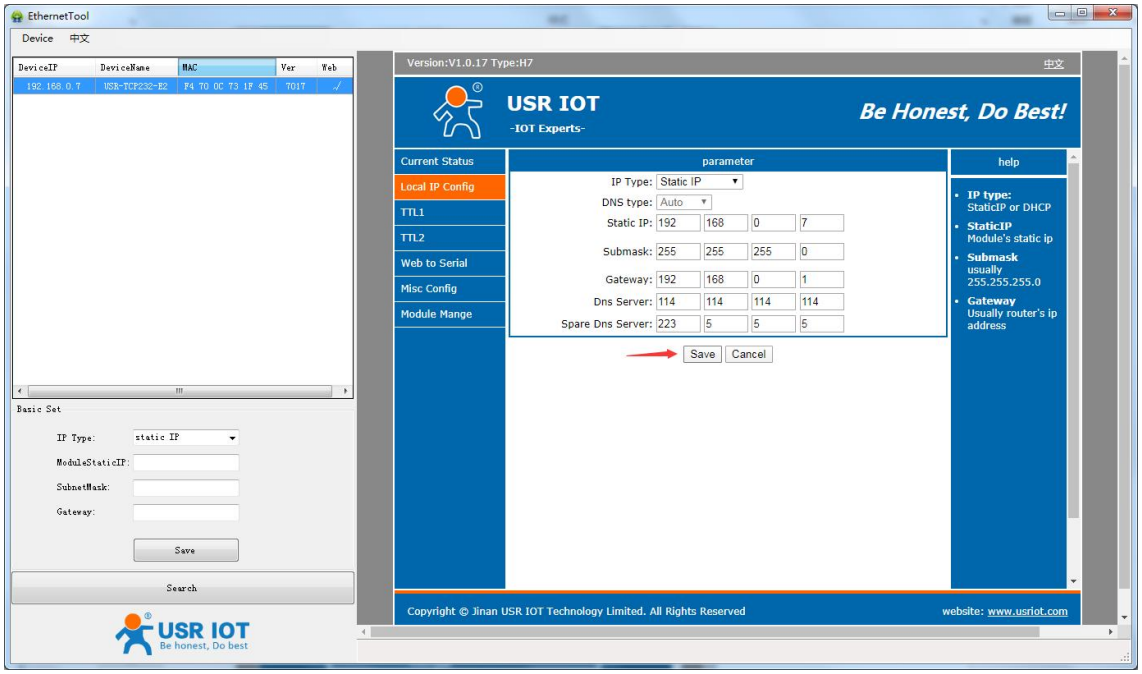
(6) Switch language to *English;*



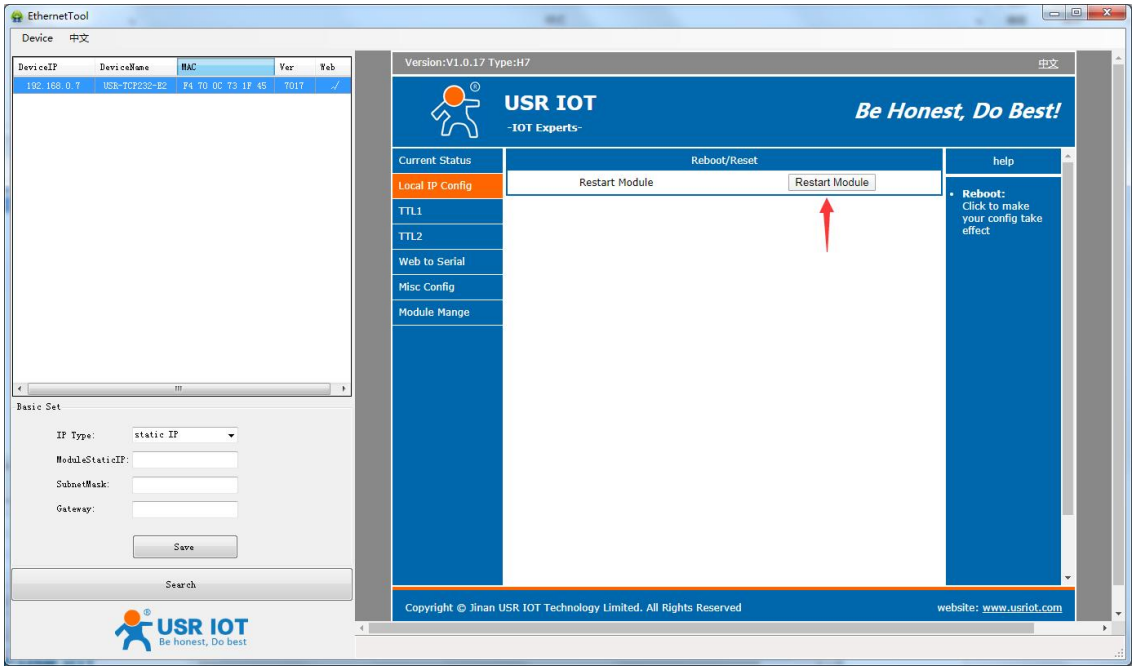
(7) Modify the *IP address and Gateway* of the BU module on this page;



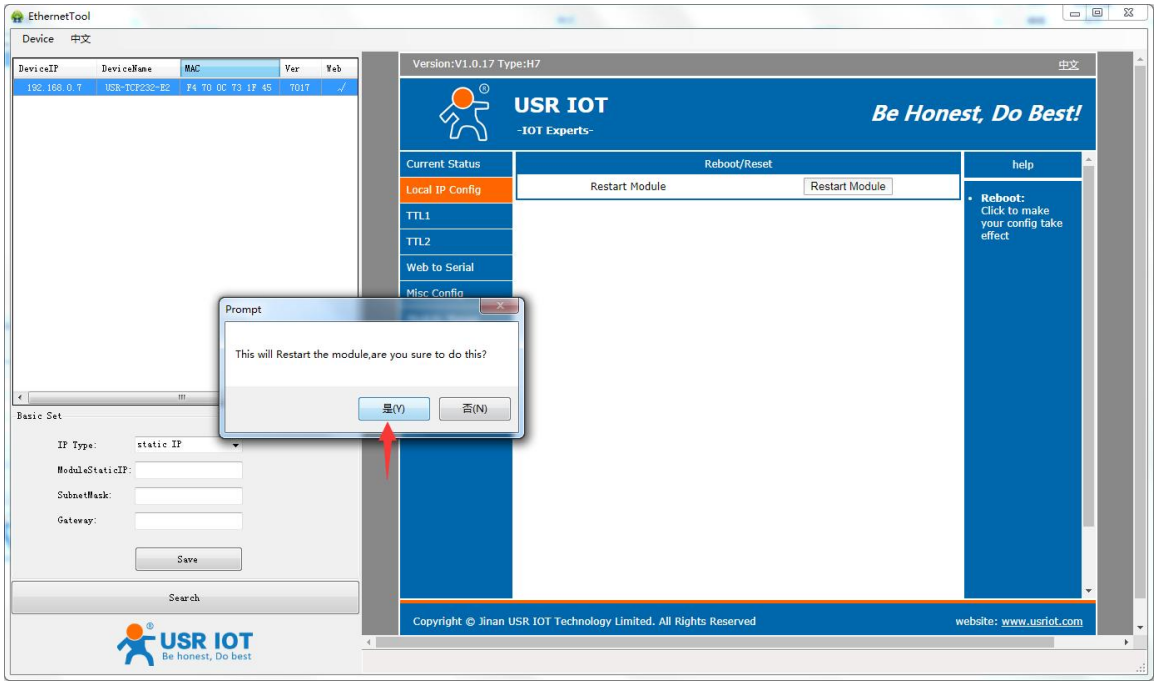
Click *Save* when the change is complete;



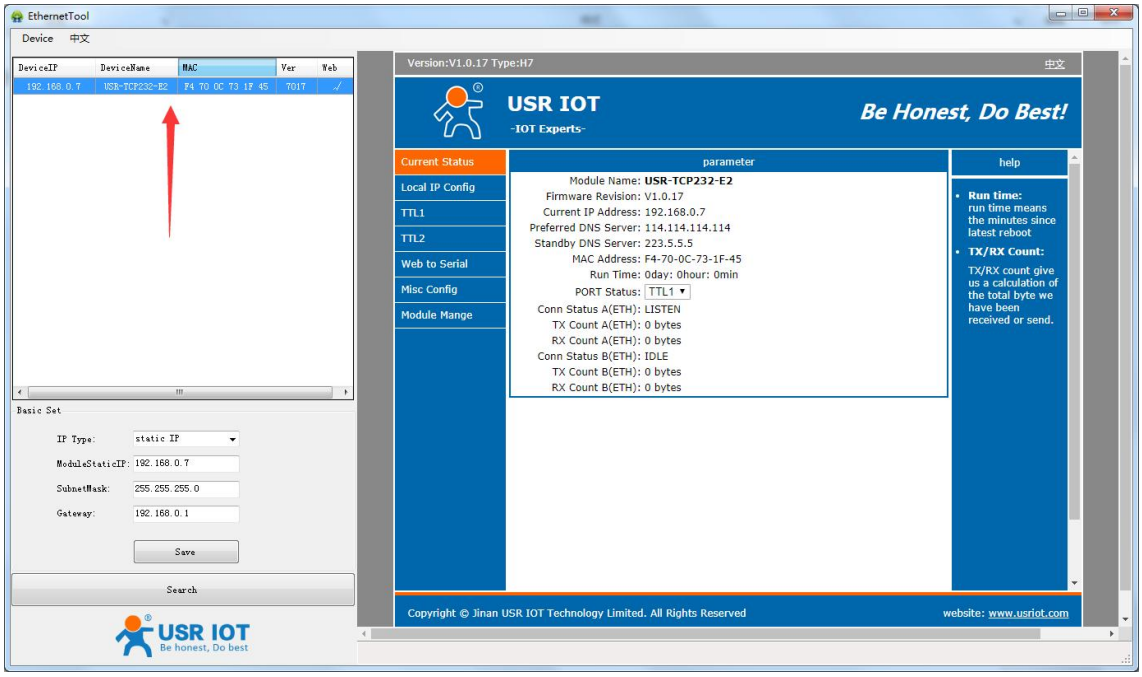
Then click *Restart Module* to allow the changes to take effect;



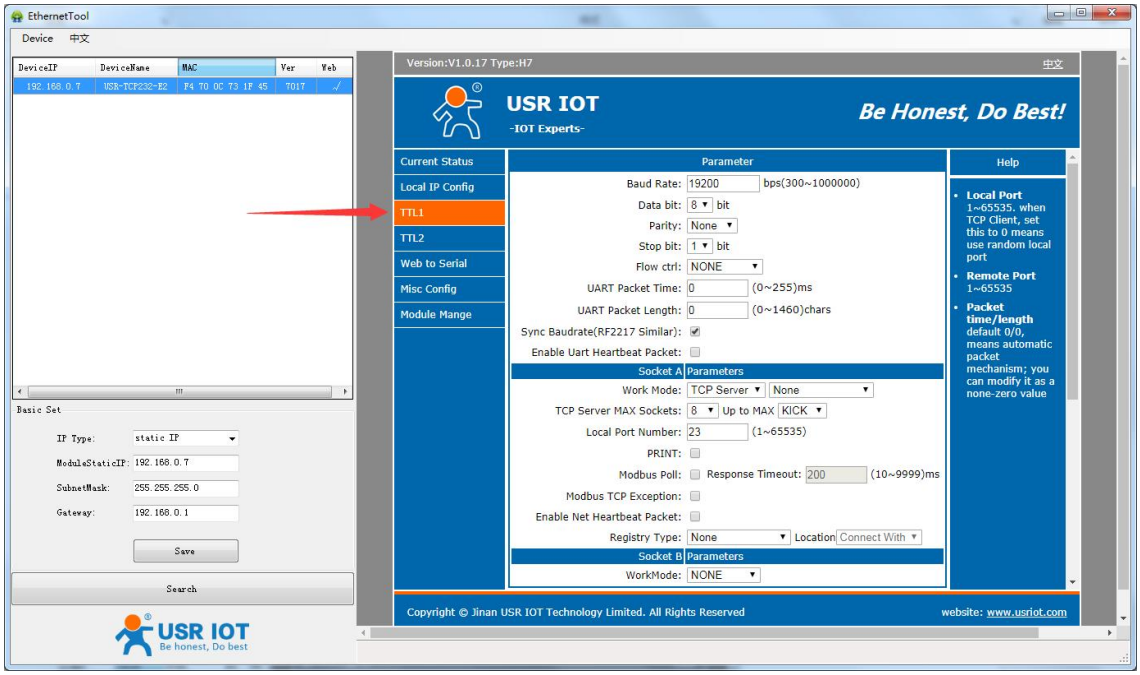
Click *Yes;*



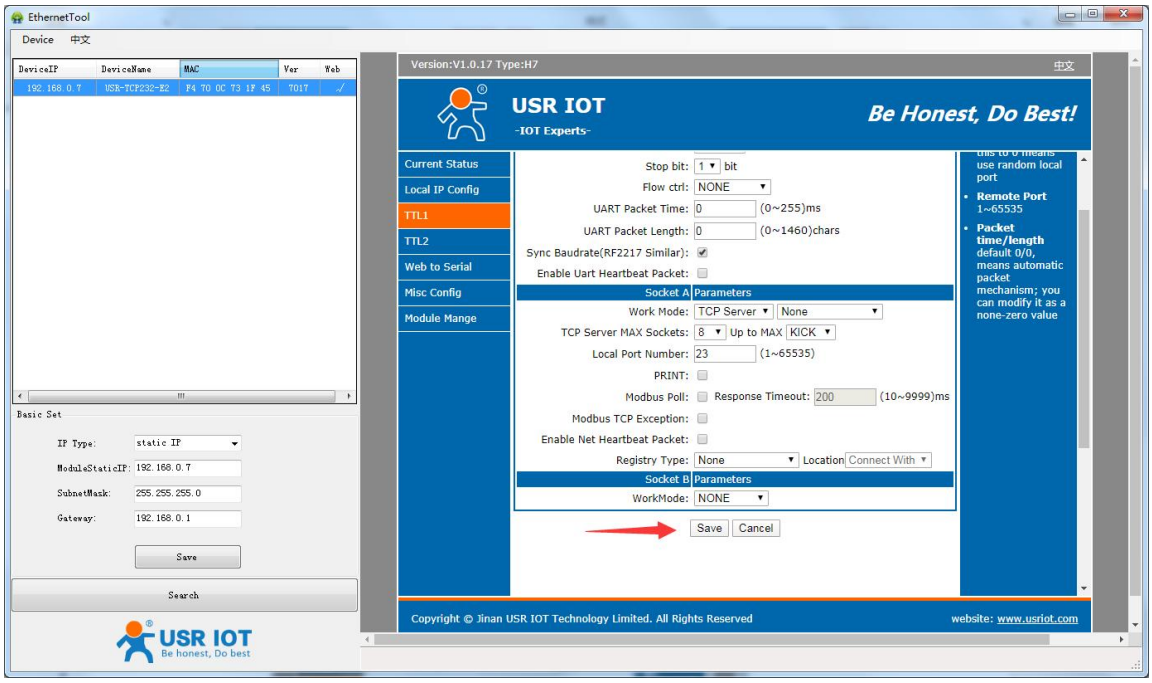
(8) Modify the B*aud Rate:* double-click the *IP address of the BU module;*



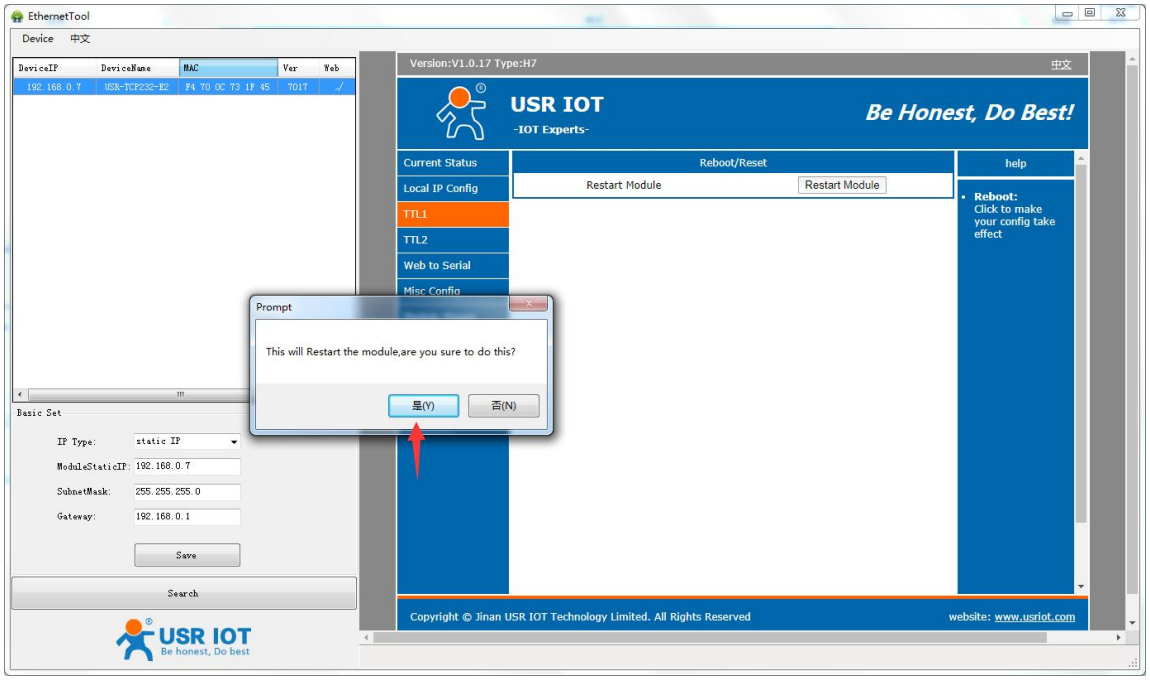
Modify the B*aud Rate* of the BU and CU control board, and the *Local Port Number* of the BU;



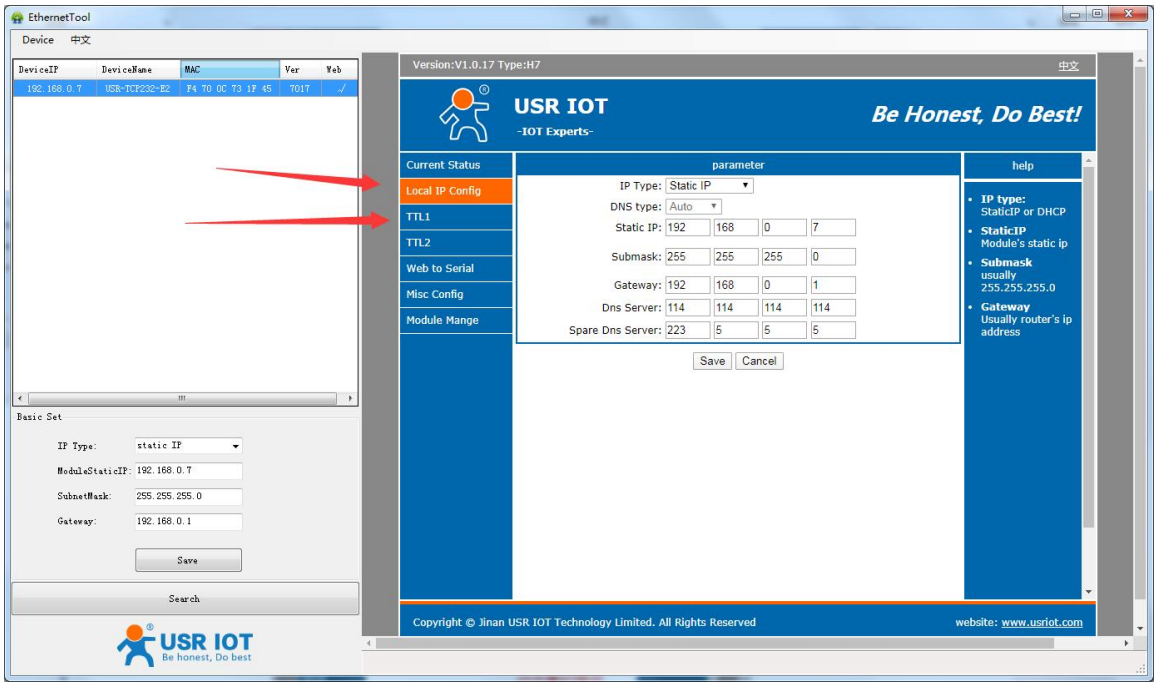
(9) Click *Save;*



(10) Click *Restart Module*, and click *Yes*;



(11) Double-click the IP address of the BU module that has been changed, open the two pages, and check whether the change takes effect.



Stage 3 Test

After setting BU, open test software “TCP&UDPDebug” ,

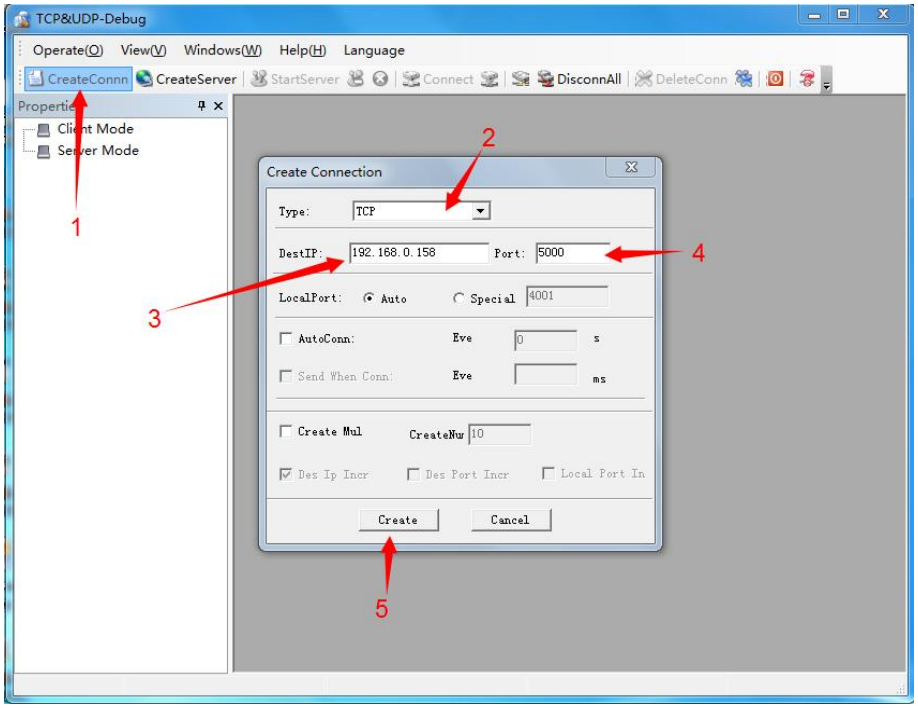
(1) click “*Create Connn*”,

(2) Type: *TCP*;

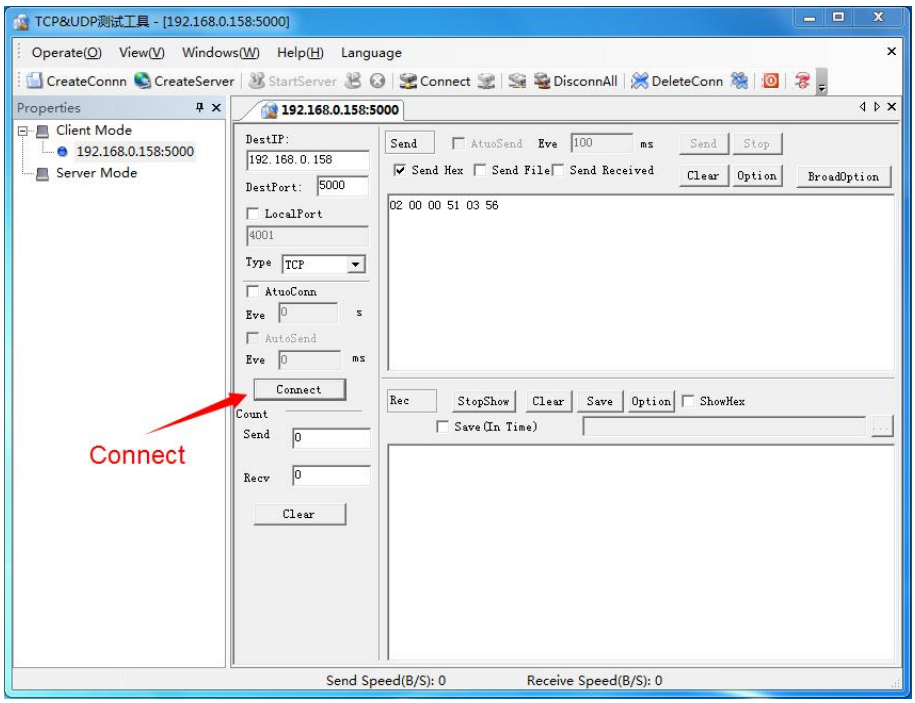
(3) DestIP: The *configured BU address* (for example, 192.168.0.7) ;

(4) Port: BU work port, which is same as “*local port number*” (for example, 5000);

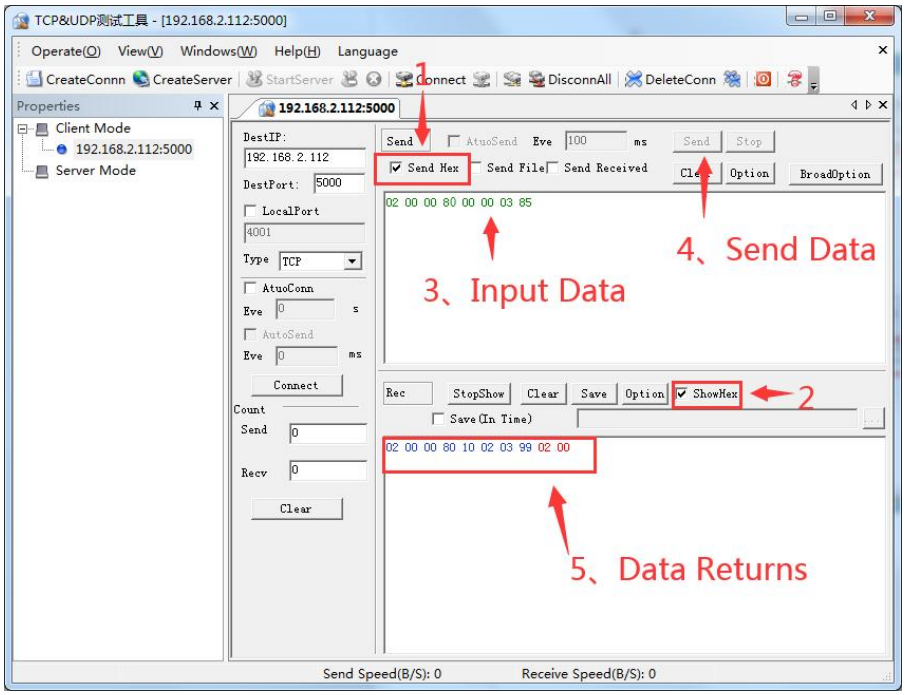
(5) Then Click *Create*; as follow picture;



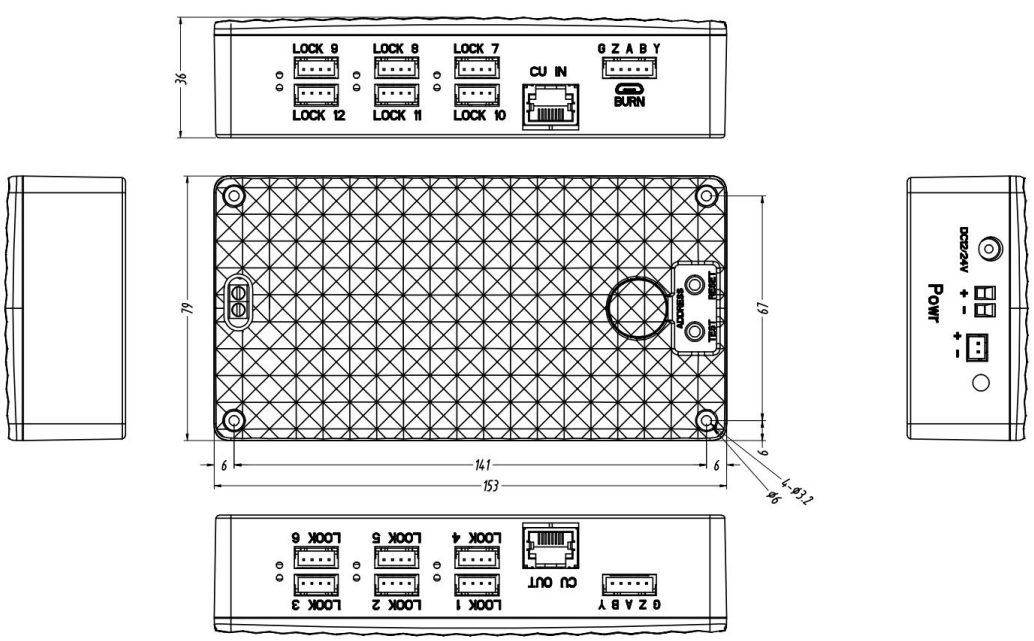
(6) Click *Connect;*



(7) Use the CU12 protocol for testing. Notice: Select *Send HEX* and *Show HEX.*



4.Installation Instruction



Laputa